## 14-2e The Firm’s Long-Run Decision to Exit or Enter a Market

A firm’s long-run decision to exit a market is similar to its shutdown decision. If the firm exits, it will again lose all revenue from the sale of its product, but now it will save not only its variable costs of production but also its fixed costs. Thus, *the firm exits the market if the revenue it would get from producing is less than its total cost of production.*

We can again make this rule more useful by writing it mathematically. If $TR$ stands for total revenue and $TC$ stands for total cost, then the firm’s exit rule can be written as

$$\text{Exit if } TR < TC.$$

The firm exits if total revenue is less than total cost. By dividing both sides of this inequality by quantity $Q$, we can write it as

$$\text{Exit if } TR/Q < TC/Q.$$

We can simplify this further by noting that $TR/Q$ is average revenue, which equals the price $P$, and that $TC/Q$ is average total cost, $ATC$. Therefore, the firm’s exit rule is

$$\text{Exit if } P < ATC.$$

That is, a firm chooses to exit if the price of its good is less than the average total cost of production.

A parallel analysis applies to an entrepreneur who is considering starting a firm. He will enter the market if starting the firm would be profitable, which occurs if the price of the good exceeds the average total cost of production. The entry rule is

$$\text{Enter if } P > ATC.$$

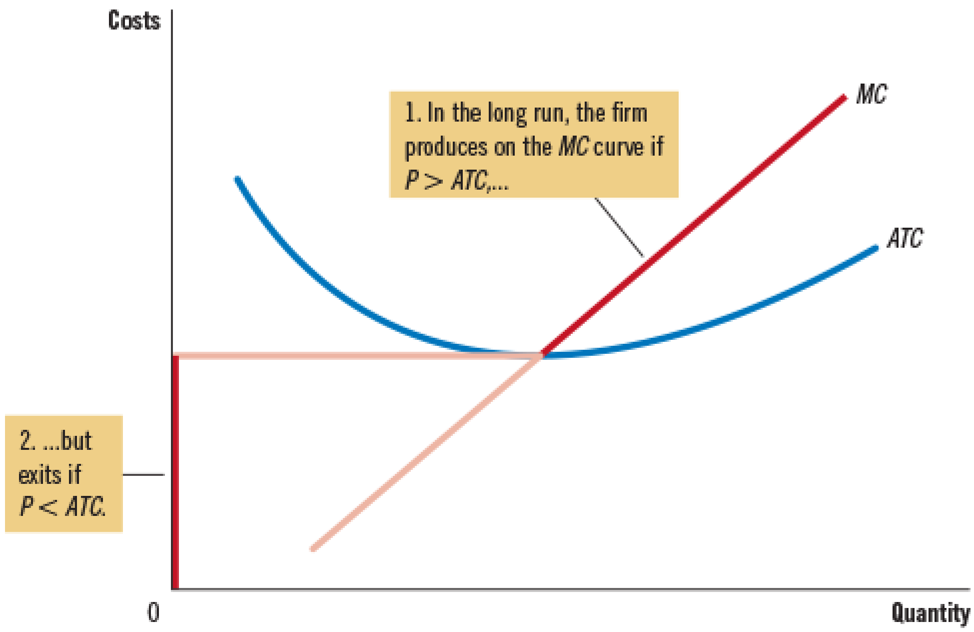
The rule for entry is exactly the opposite of the rule for exit.

We can now describe a competitive firm’s long-run profit-maximizing strategy. If the firm produces anything, it chooses the quantity at which marginal cost equals the price of the good. Yet if the price is less than the average total cost at that quantity, the firm chooses to exit (or not enter) the market. These results are illustrated in Figure 4. *The competitive firm’s long-run supply curve is the portion of its marginal-cost curve that lies above the average-total-cost curve.*

**Figure 4**

**The Competitive Firm’s Long-Run Supply Curve**

In the long run, the competitive firm’s supply curve is the portion of its marginal-cost curve ($MC$) that lies above its average-total-cost curve ($ATC$). If the price falls below average total cost, the firm is better off exiting the market.




Chapter 14: Firms in Competitive Markets: 14-2e The Firm’s Long-Run Decision to Exit or Enter a Market
Book Title: WGU Economics
Printed By: Mike Benton (mbento6@wgu.edu)